Coding & Solution

Date	2 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Durga1607/credit-card-approval-prediction
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	Data preprocessing, feature engineering, EDA, Logistic Regression, Decision Tree, Random Forest, XGBoost model training, model evaluation, Flask web application, real-time credit card approval prediction, IBM Cloud deployment support
Pending / Incomplete Features	Advanced user authentication, database integration, explainable AI dashboard, email notification system
Setup / Run Instructions	Install required Python packages, place the dataset in the project directory, run the training notebook to generate best_model.pkl, then execute python app.py and open the Flask application in a web browser

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture by separating data preprocessing, feature engineering, model training, evaluation, and Flask application logic into independent components. Multiple machine learning algorithms are compared to select the best-performing model for deployment. The source code is organized for readability, maintainability, and future enhancements while supporting real-time credit card approval prediction.